Test Plan

SauceDemo Web Application

Manual UI and UI Automation Testing

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1. Introduction

This Test Plan describes the planned testing approach for the SauceDemo web application. SauceDemo is a demo e-commerce website designed to simulate common online shopping functionalities such as user authentication, product browsing, cart management, and checkout.

The purpose of this Test Plan is to define the scope, objectives, test approach, environment, and planned scenarios for manual UI testing and UI automation testing.

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2. Test Objectives

The objectives of this testing effort are to:

- Verify that the main functionalities of the SauceDemo application work as expected
- Validate user flows based on the observable behavior of the SauceDemo website
- Plan positive (happy path) and negative test scenarios
- Identify potential functional and validation issues through manual UI testing
- Define selected functional UI scenarios for automation testing

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3. Test Scope

3.1 In Scope

The following features and activities are planned to be tested:

- User login and logout
- Display of product list
- Product sorting functionality
- Adding items to the shopping cart
- Removing items from the shopping cart
- Viewing cart contents

- Checkout process
- Checkout form validation
- Order completion confirmation
- Basic observation of UI response time during user actions (manual observation only)
- UI automation testing using Selenium WebDriver

3.2 Out of Scope

The following activities are not included in this test plan:

- Tool-based performance testing (load, stress, endurance)
- Security testing
- Compatibility testing on multiple browsers or devices
- Backend or database testing
- API testing

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4. Test Types

Manual Testing

- Functional testing
- Positive (happy path) testing
- Negative testing
- UI validation testing
- Manual observation of UI responsiveness

Automation Testing

- Functional UI automation testing
- Repeated execution of selected scenarios
- End-to-end user flow automation

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5. Test Environment

Item	Description
Application URL	https://www.saucedemo.com
Browser	Safari
Operating System	macOS

Automation Tool	Selenium WebDriver
Programming Language	Java
Test Framework	TestNG
Build Tool	Maven

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6. Test Data

The following test credentials provided by SauceDemo are planned to be used during testing.

Accepted usernames are:

- standard_user
- locked_out_user
- problem_user
- performance_glitch_user
- error_user
- visual_user

Password for all users:

- secret_sauce

For negative login testing, invalid credentials will be used.

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7. Entry and Exit Criteria

Entry Criteria

- SauceDemo website is accessible
- Test environment is ready
- Test scenarios and test cases are prepared

Exit Criteria

- All planned test scenarios are executed
- Test results and execution evidence are documented
- Selected UI automation scripts are executed successfully

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8. Risks and Assumptions

Risks

- Temporary SauceDemo website unavailability
- UI changes that may affect automation locators

Assumptions

- Testing will be based solely on the visible behavior of the SauceDemo application
- No formal requirements document is provided

8.1Test Scenarios (Path-Based – End-to-End)

Feature 1: Login Functionality

Scenario 1 – Happy Path: Successful Login
A user logs in with valid credentials and accesses the inventory page.

Scenario 2 – Negative Path: Invalid Login
A user attempts to log in with invalid credentials and is denied access to the system.

Scenario 3 – Negative Path: Locked User Login
A locked user attempts to log in and is prevented from accessing the application.

Feature 2: Product Listing & Sorting

Scenario 4 – Happy Path: View Product List
A logged-in user views the product inventory after successful authentication.

Scenario 5 – Happy Path: Sort Products by Price
A logged-in user sorts products by price from low to high and views the updated list.

Feature 3: Cart Management

Scenario 6 – Happy Path: Add and Remove Products from Cart
A logged-in user adds products to the cart, removes a product, and verifies the cart is updated.

Feature 4: Checkout Process

Scenario 7 – Happy Path: Complete Checkout Successfully
A logged-in user adds products to the cart, enters valid checkout information, completes the checkout, and receives order confirmation.

Scenario 8 – Negative Path: Checkout with Missing Information
A logged-in user attempts to complete checkout with missing required fields and receives a validation error.

Feature 5: Logout & Session Handling

Scenario 9 – Happy Path: Successful Logout
A logged-in user logs out successfully and is returned to the login page.

Scenario 10 – Negative Path: Access After Logout
A logged-out user attempts to access a protected page and is redirected to the login page.

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9. Automation Coverage

The following scenarios are planned to be automated using Selenium WebDriver with Java and TestNG:

- Login with valid credentials
- Login with invalid credentials
- Product sorting by price
- Add item to cart
- Remove item from cart
- Checkout with valid information
- Checkout validation for missing required fields
- Logout functionality

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10. Traceability Matrix

Feature	Scenario ID	Test Type
Login	Scenario 1–3	Manual and Automation
Products	Scenario 4–5	Manual and Automation
Cart	Scenario 6–8	Manual and Automation
Checkout	Scenario 9–10	Manual and Automation

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11. Defect Management

Defects identified during testing will be documented using an Excel-based defect log.

Each defect record will include:

- Defect ID
- Summary
- Steps to reproduce
- Expected result
- Actual result
- Severity and priority
- Evidence (screenshots or logs)

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12. Schedule and Resources

Resources

- One QA engineer responsible for manual testing and UI automation

Tentative Schedule

- Day 1: Test planning and environment setup
- Day 2–3: Manual test execution
- Day 4: UI automation development
- Day 5: Retesting and reporting

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13. Conclusion

This Test Plan defines the planned approach for testing the SauceDemo web application using manual UI testing and UI automation testing. The plan establishes scope, scenarios, and testing strategy prior to execution.